Plan of action

The Beast Locker

Creation of the eCommerce platform

Study: INF Internship 25/26

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2026-02-12

Version 1.0

Version Control

Version	Date	Changes	Initials
0.5	2026-02-12	Initial version	MA
1.0	2026-02-18	Finalised Gantt chart planning	MA

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Abbreviations

TBL	The Beast Locker

1. Introduction

1.1 Background Information

As part of my Software Engineering and Business Informatics internship at Fontys University of Applied Sciences, I will design and develop an eCommerce platform titled *The Beast Locker*. The project focuses on creating a modern web application that supports online product presentation and ordering for a real stakeholder.

1.2 Company Description

The internship takes place at Beast Team MMA SRL, a sports organization based in Râmnicu Vâlcea, Romania. The company provides training in contact sports such as MMA and Brazilian Jiu-Jitsu and is currently expanding its brand by developing its own sports equipment line. To support this expansion, the company aims to strengthen its digital presence through a dedicated online platform.

1.3 Project Context

Currently, the company lacks a structured eCommerce solution tailored to its needs. My role is to design and implement a responsive web platform that allows users to browse products and place orders while applying modern UX principles and web development practices. Payment handling and logistics integration are outside the project scope and will be implemented later through third-party services.

2. Assignment Overview

2.1 Current Situation

The Beast Team currently promotes its activities mainly through a general website and social media. Although the company is developing its own sports equipment brand, it does not have a dedicated platform for structured online product presentation or ordering. This limits scalability and reduces opportunities for national exposure.

2.2 Problem Outline

The absence of a dedicated eCommerce platform prevents the company from effectively promoting and selling products online. Without such a system, product presentation is inconsistent, online visibility remains limited, and expansion beyond local audiences becomes difficult.

2.3 Proposed Solution

To address this problem, I will develop *The Beast Locker*, a responsive eCommerce platform designed specifically for the company's needs. The system will include structured product presentation, ordering functionality, and a modern user interface based on UX and design research. The project includes analysis, design, implementation, and testing phases.

2.4 Expected Outcomes

The expected outcome is a functional web platform that supports online product sales and improves brand visibility. The project will provide a scalable technical foundation for future digital growth while allowing me to develop skills in web development, UX design, and stakeholder collaboration.

Minimum Viable Product (MVP)

The minimum viable product for this project consists of a functional and responsive eCommerce platform that enables users to browse products, view detailed product information, and complete the ordering process through a clear and intuitive interface. The MVP focuses on implementing the core "Must Have" features identified in the MoSCoW prioritization, ensuring usability, stability, and a scalable foundation for future enhancements.

3. Project overview and planning

3.1 Project Scope

In scope:

- Design and development of a responsive eCommerce platform.
- Product catalogue, product pages, user account system and ordering flow.
- UX design, research, testing and SEO-oriented implementation.

Out of scope:

- Payment gateway integration.
- Logistics and delivery management.
- Long-term maintenance after internship completion.

Clearly defining the project scope helps maintain the alignment between stakeholder expectations and the limited internship timeframe. By explicitly identifying excluded features such as payment gateway integration and logistics systems, I can keep focus on delivering a stable and high-quality minimum viable product within realistic constraints.

3.2 Stakeholders

The project involves multiple stakeholders who contribute technical guidance, business direction, academic supervision, and user perspective.

- Mehtupciu Alexandru – Developer (intern).
- Toma Lucian – Technical supervisor and mentor.
- Alexandru Codita – CEO and business stakeholder.
- Fontys supervisor (Gé Schriever) – Academic supervision and evaluation.
- End users – Customers interacting with the platform.

3.3 Methodology

The project follows an Agile-inspired iterative development approach. Work is divided into repeated cycles consisting of research, UX design, implementation, testing, and stakeholder feedback. Each iteration allows refinement of requirements and continuous improvement of usability and technical quality while maintaining focus on delivering the defined minimum viable product. The development workflow is illustrated through a UML-style activity diagram provided in Appendix A.

3.4 Research Approach

Main research question
How can I design and implement an eCommerce platform that effectively presents the company's sports equipment brand while applying modern UX principles and SEO practices to support online visibility and usability?

Sub research questions

- What UX design principles are most effective for presenting products in small-to-medium eCommerce platforms?
- What SEO guidelines should be applied during development to improve search visibility without paid advertising?
- What technical and design approaches best support responsive performance and usability across devices?

Research Methods
The research will primarily consist of desk research, including analysis of competitor platforms, UX guidelines, and established SEO best practices. Additionally, iterative experimentation will be applied during development, where design and technical decisions are evaluated through practical testing and refinement. This trial-and-error approach supports continuous improvement while remaining aligned with Agile development principles.

3.5 Feature Prioritization

Must Have	Should Have	Could Have	Will Not Have
Responsive web design	Advanced filtering options	Animations or micro-interactions	Custom payment gateway integration
Product catalogue	SEO optimization improvements	Wishlist functionality	Logistics management system
Product detail pages	Enhanced product visuals	Advanced analytics dashboard	Mobile native application
User account system	Performance optimizations	Personalization features	Inventory automation system
Shopping cart and ordering	Basic accessibility improvements	Marketing integrations	Multi-language support (initial phase)
Clear navigation and UI	Image optimization	Additional UI animations	External ERP integrations

3.6 Deliverables, Definition of Done and Success Criteria

Project Deliverables

- Research documentation and design artifacts (wireframes, UX decisions, research documents).
- Functional eCommerce web platform (*The Beast Locker*).

- Product database with structured product presentation.
- Technical documentation describing architecture and implementation decisions.

Definition of Done

A feature is considered complete when it is fully implemented, tested practically, and is not missing any functionality (e.g.: A menu bar is considered done when all of the sections exist, have been manually tested in terms of functionality, usability and responsiveness, and there are no placeholders left). The platform should operate reliably, provide responsive design across devices, and support a clear user flow for browsing and ordering products.

Project Success Criteria
The project will be considered successful if:

- A functional and responsive eCommerce platform is delivered.
- Core features from the "Must Have" category are implemented.
- The platform improves product presentation and brand visibility.
- The system demonstrates usability and performance suitable for real users.

3.7 Risks

Several risks may affect the execution of the project:

- **Changing requirements**: Stakeholder expectations may evolve during development. Mitigation: maintain regular communication and iterative validation of features.
- **Technical challenges**: Unexpected implementation difficulties may occur. Mitigation: prioritize core functionality and apply incremental development.
- **Time constraints**: Limited internship duration may restrict feature scope. Mitigation: focus on delivering a minimum viable product based on the "Must Have" features.

3.8 Planning

To ensure the project remains feasible within the internship timeframe, I planned the work in weekly phases. The planning below reflects my current progress and the remaining steps toward delivery. The first weeks focus on onboarding and tooling, followed by research and UX design, then implementation and testing. The final weeks are reserved for deployment, final reporting, and stakeholder-driven adjustments.

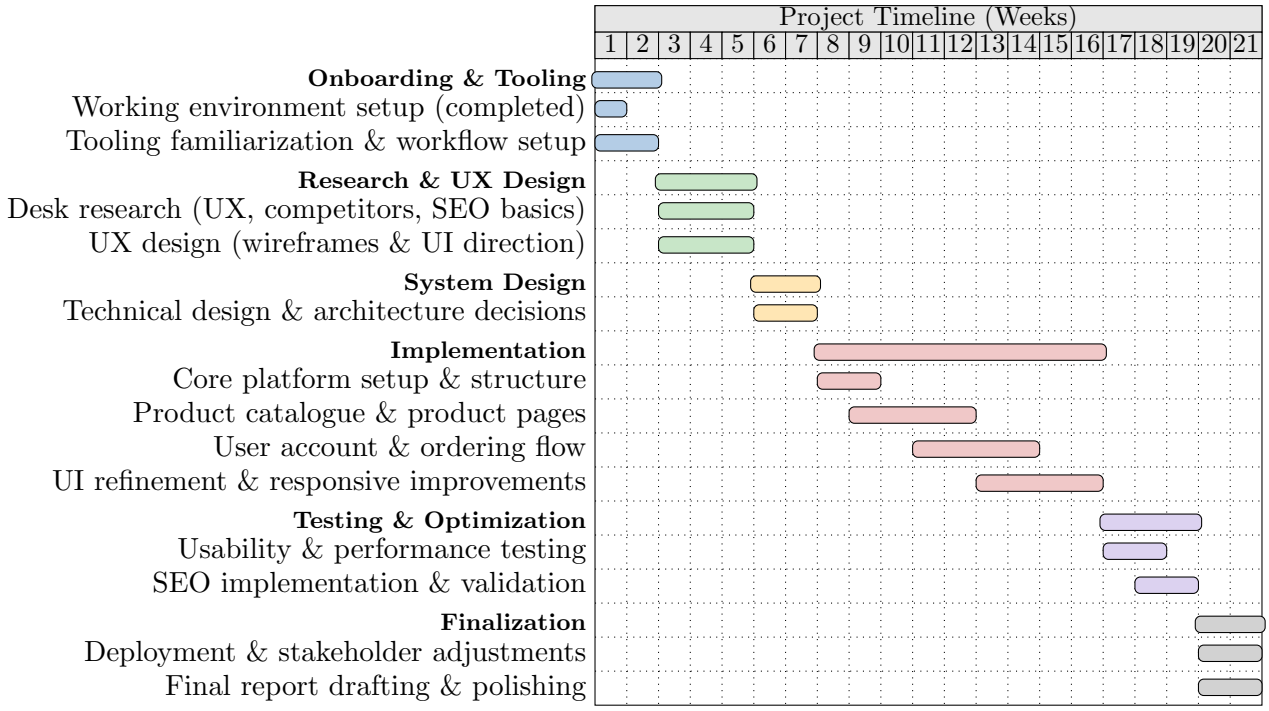

Figure 1: High-level project planning overview

3.9 Constraints and Assumptions

Since this project is to be completed within the timeframe of the internship, I need to work within certain practical constraints. The limited duration means that prioritization is essential, which is why the focus remains on delivering a stable minimum viable product rather than attempting to implement every possible feature. I also rely on regular stakeholder availability for feedback and validation, as this helps guide decisions and ensures that development stays aligned with real business needs. However, in writing this plan I also realise the importance of innovation and independance. Working in a real professional environment means adapting and coming up with solutions myself when nobody is present to give feedback or guidance.
It is to be assumed that product content, branding materials, and business requirements will be provided by the stakeholders when needed. Additionally, deployment infrastructure is expected to be available toward the end of the internship to support final delivery. These assumptions help me plan realistically while leaving room for iterative improvement throughout the project.

4. Appendix

4.1 UML Activity Diagram – Development Methodology

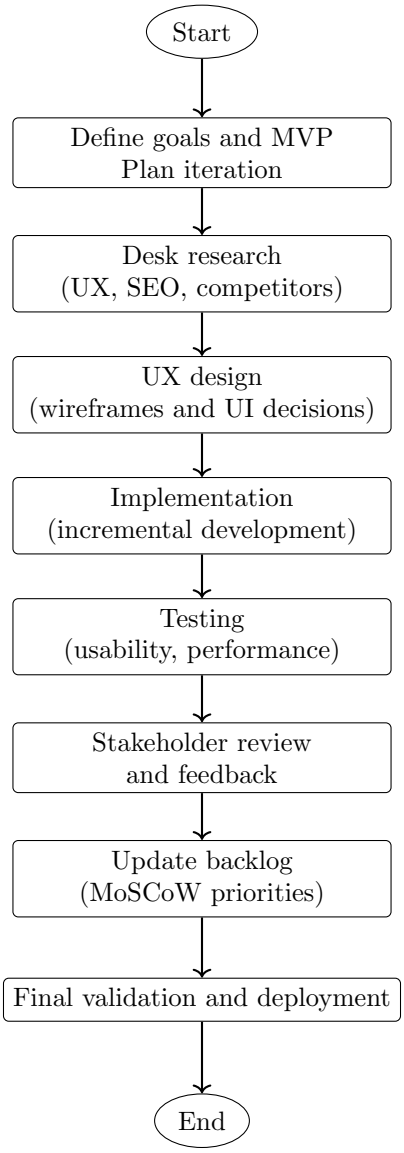

Figure 2: UML-style activity diagram representing the iterative development methodology